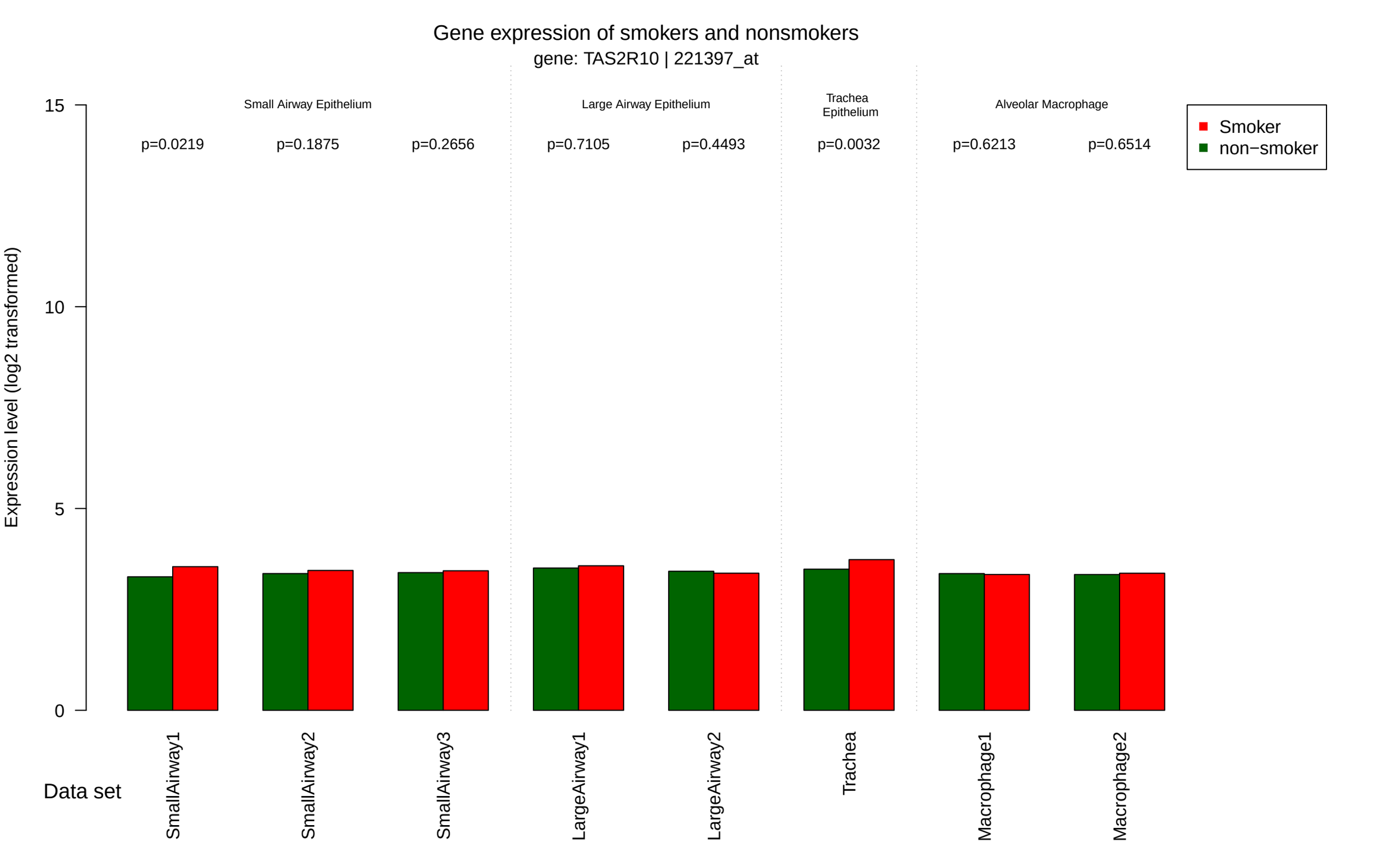
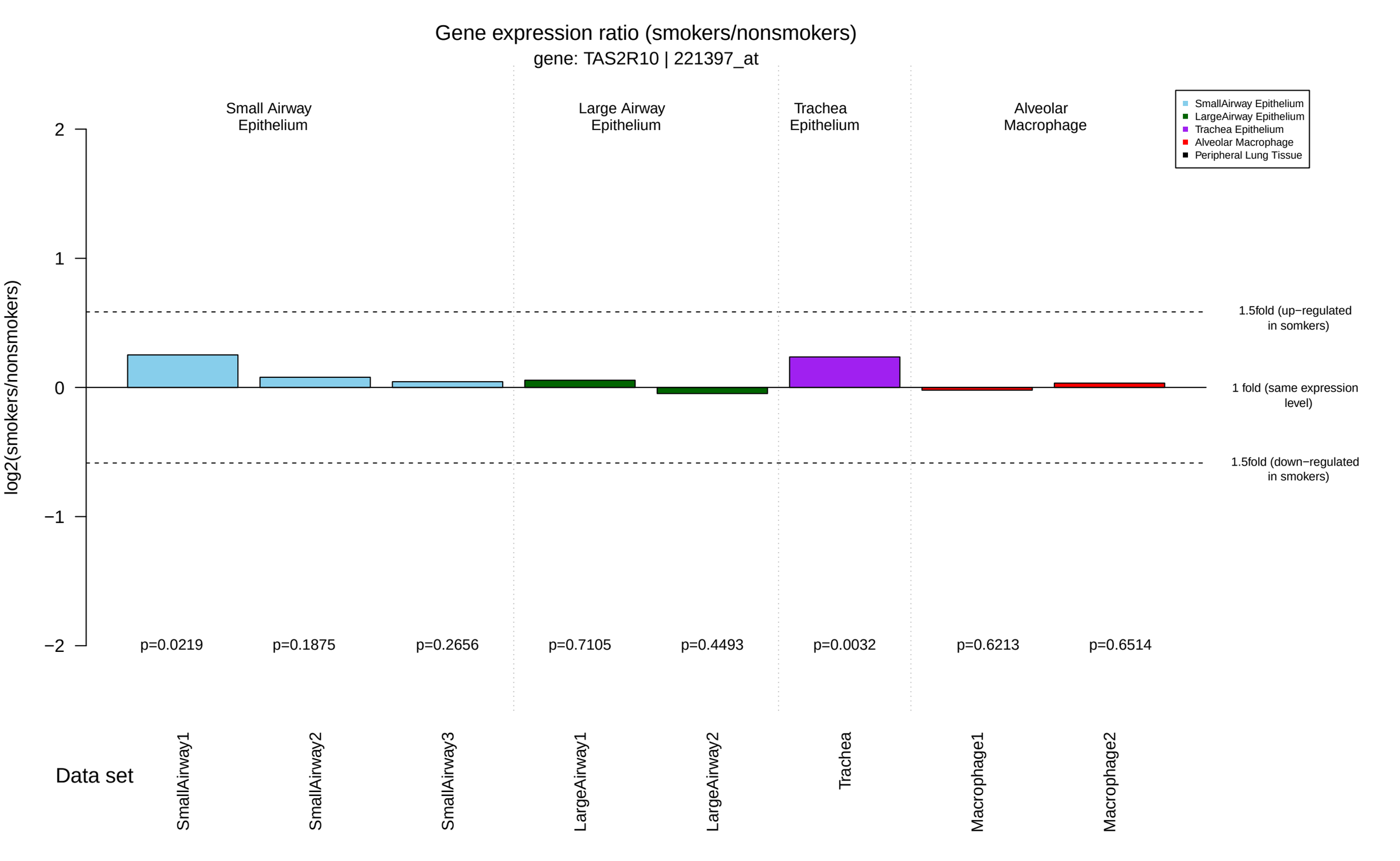
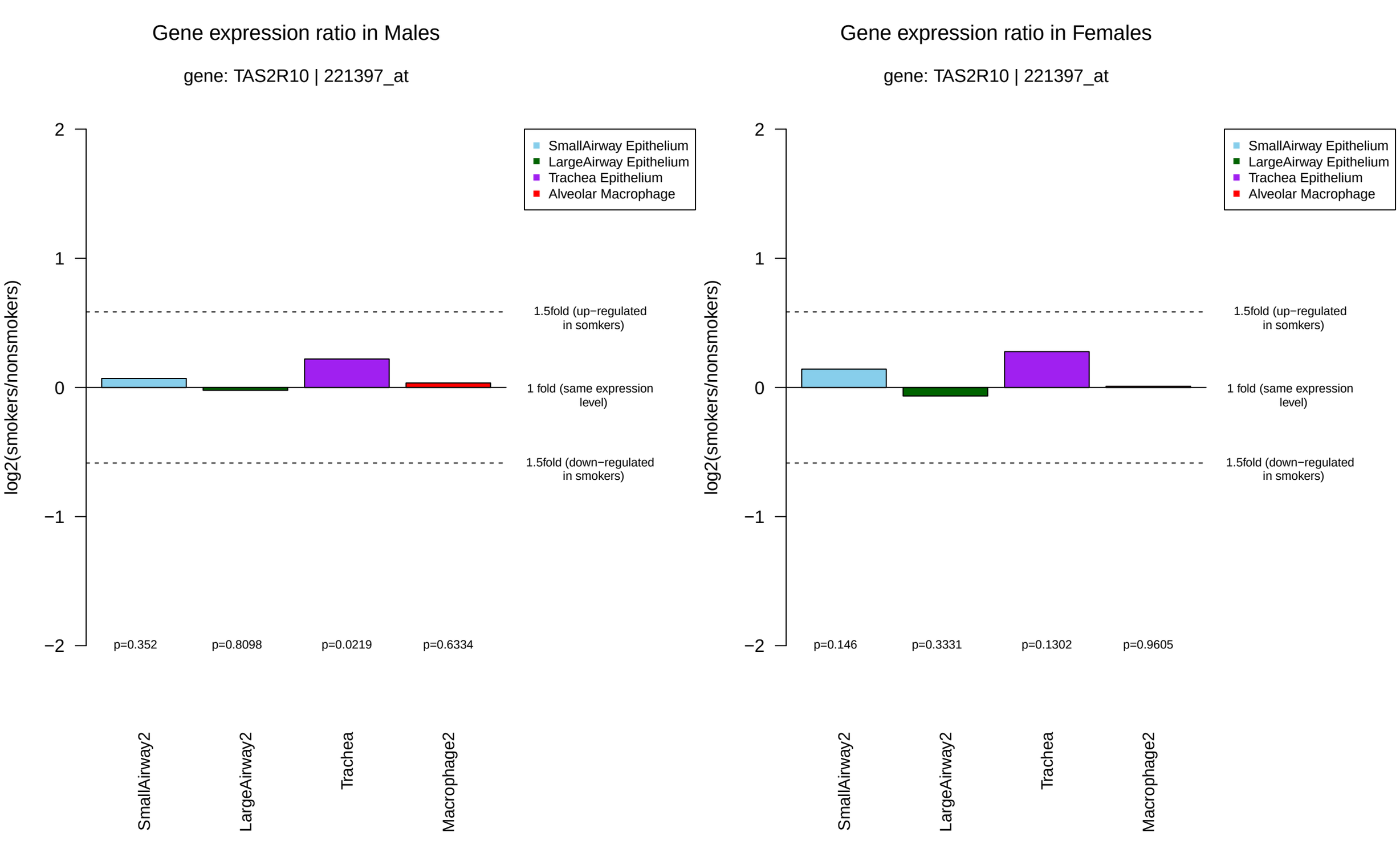
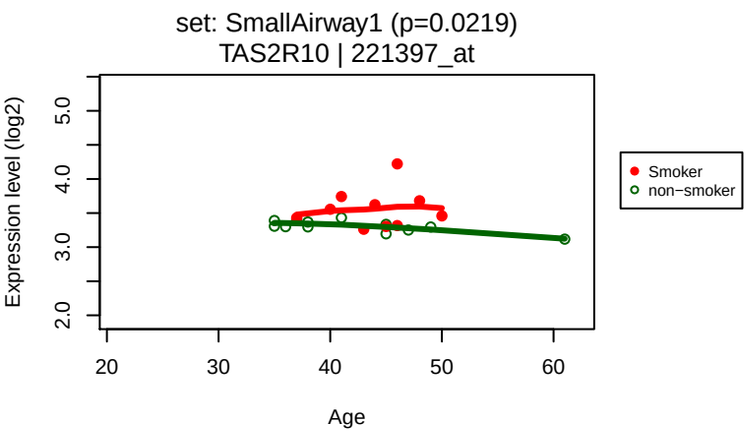
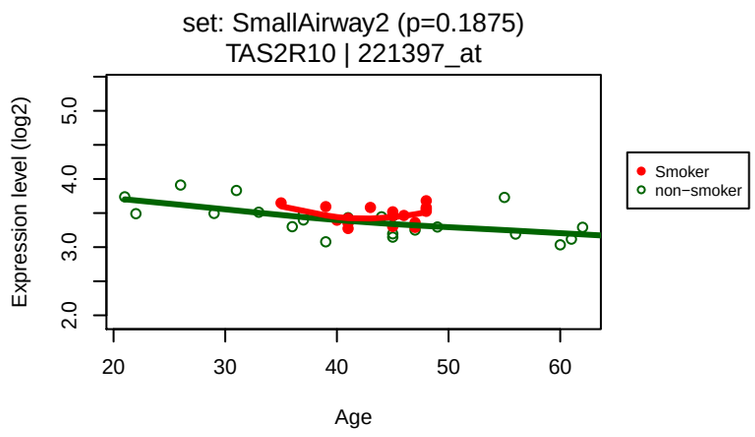
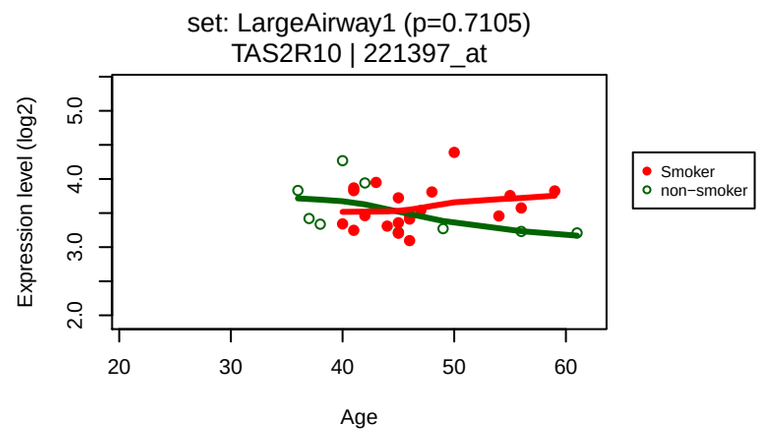

Gene expression of smokers and nonsmokers
gene: TAS2R10 | 221397_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0219
p=0.1875
p=0.2656
p=0.7105
p=0.4493
p=0.0032
p=0.6213
p=0.6514
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

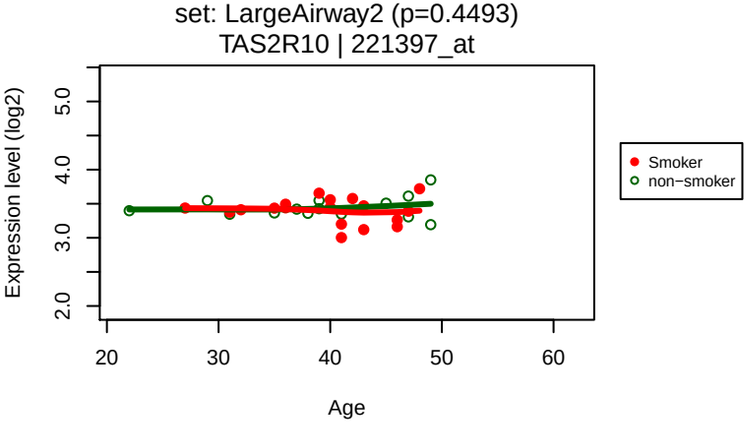

Gene expression ratio (smokers/nonsmokers)
gene: TAS2R10 | 221397_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0219
p=0.1875
p=0.2656
p=0.7105
p=0.4493
p=0.0032
p=0.6213
p=0.6514
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

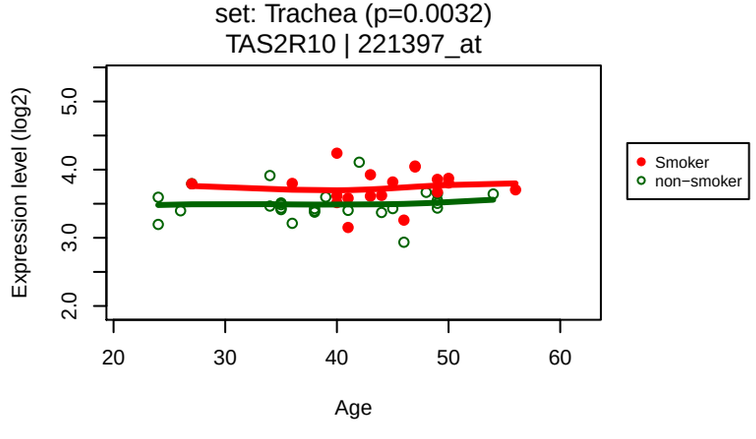

Gene expression ratio in Males
gene: TAS2R10 | 221397_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.352
p=0.8098
p=0.0219
p=0.6334
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAS2R10 | 221397_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.146
p=0.3331
p=0.1302
p=0.9605
SmallAirway2
LargeAirway2
Trachea
Macrophage2

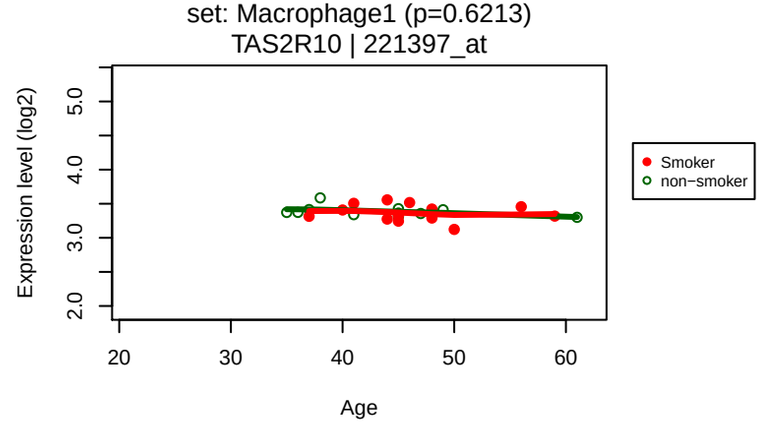

set: SmallAirway1 (p=0.0219)
TAS2R10 | 221397_at
Expression level (log2)
Age
Smoker
non−smoker

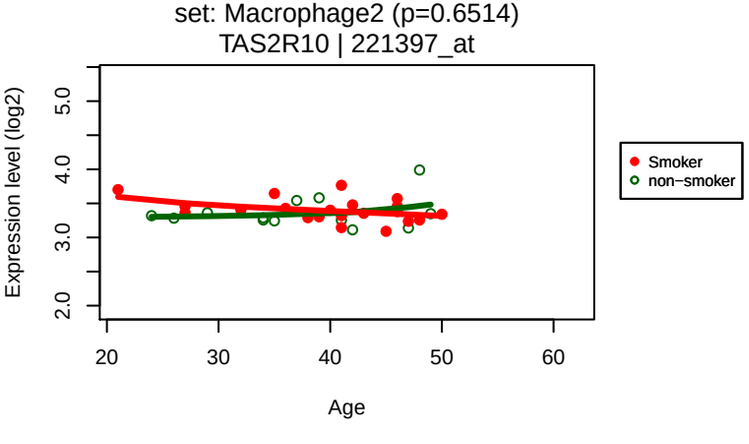

set: SmallAirway2 (p=0.1875)
TAS2R10 | 221397_at
Expression level (log2)
Age
Smoker
non−smoker

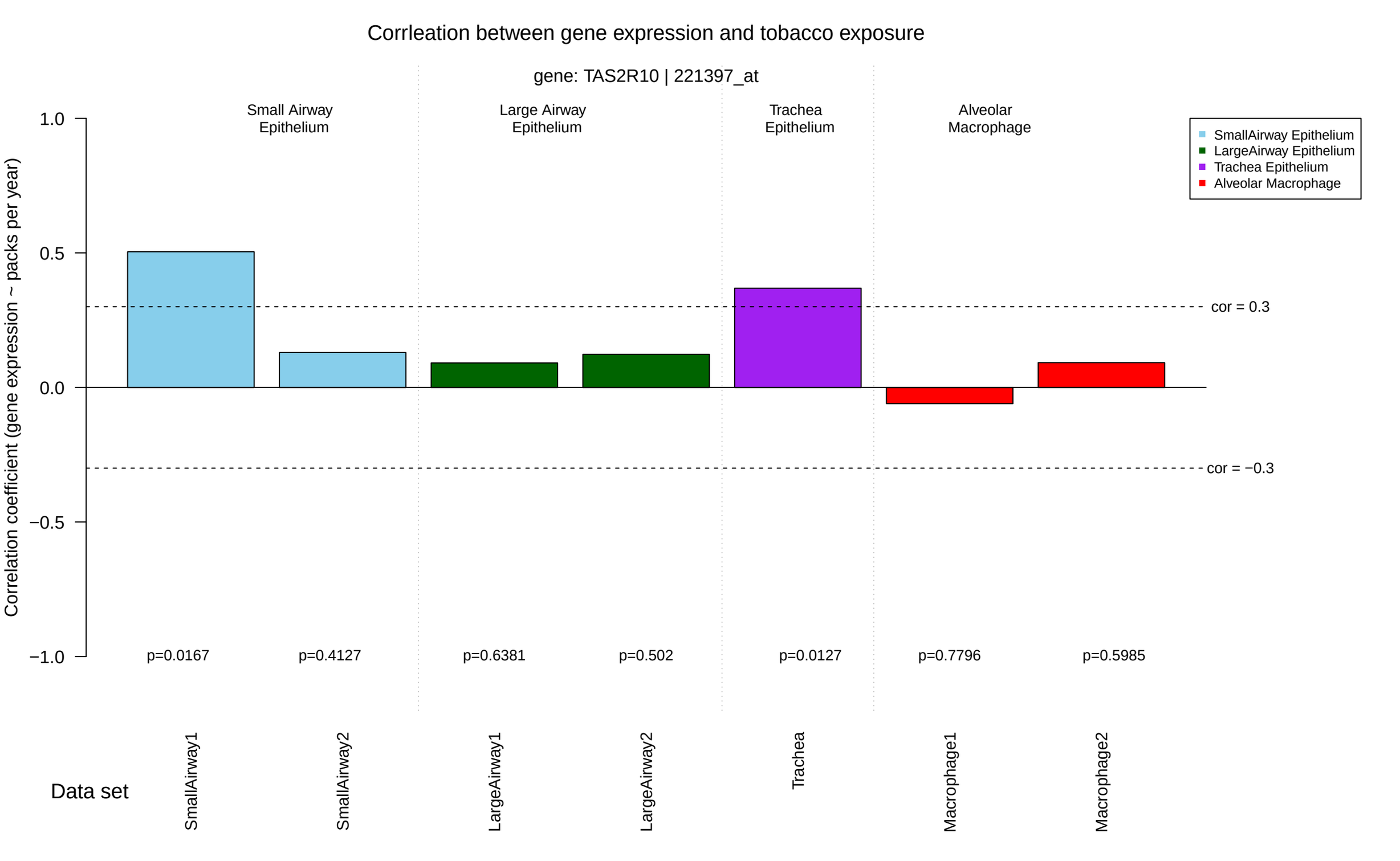

set: LargeAirway1 (p=0.7105)
TAS2R10 | 221397_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4493)
TAS2R10 | 221397_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0032)
TAS2R10 | 221397_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6213)
TAS2R10 | 221397_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6514)
TAS2R10 | 221397_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAS2R10 | 221397_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0167
p=0.4127
p=0.6381
p=0.502
p=0.0127
p=0.7796
p=0.5985
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2